

Protein

Sources of Protein

Nuts and Seeds



Dairy Products



Buzzle.com

Fish and Seafood

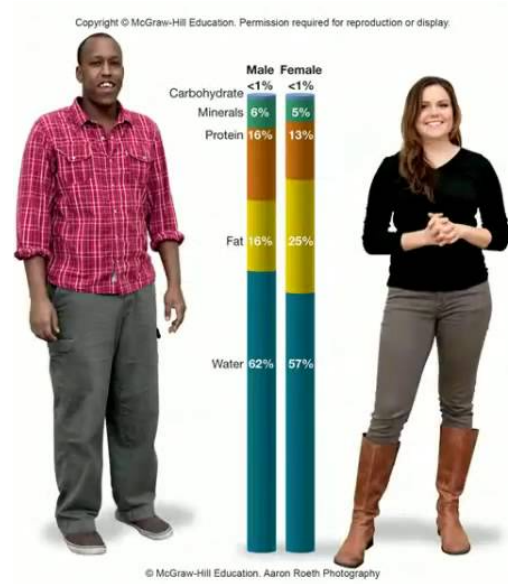


Dr. P. Satpati,
BSBE, IIT Guwahati₁

Three Misconceptions about proteins

1. Protein is the best to eat

- Carb, fats are equally important



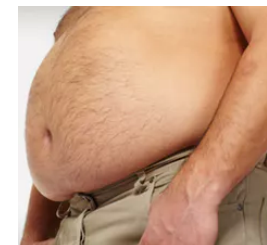
2. Protein is “animal”

- Plants are good source too

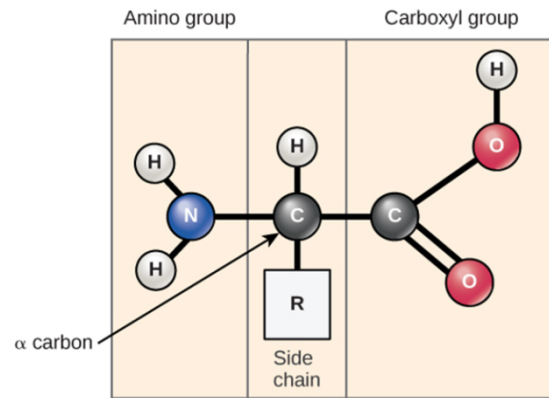
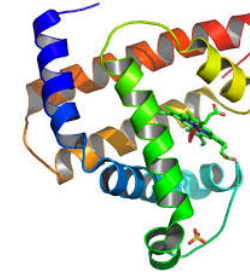
	Spinach 49% protein		Kale 45% protein		Broccoli 45% protein
	Cauliflower 40% protein		Mushrooms 38% protein		Parsley 34% protein
	Cucumbers 24% protein		Green Pepper 22% protein		Cabbage 22% protein
	Tomatoes 18% protein				

2. Eat protein and build your muscle

- Exercise does. If not you end up accumulating fat

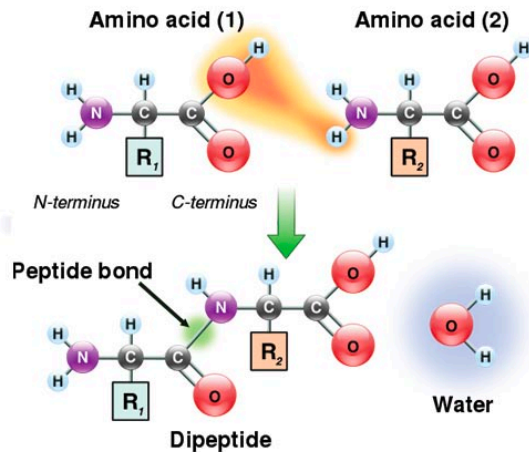
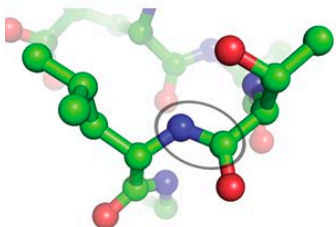


Proteins are one or multiple chains of amino acid residues.

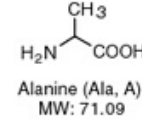
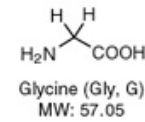


20 amino acid

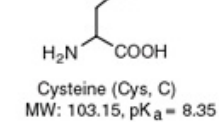
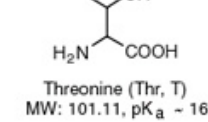
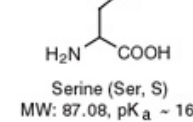
Peptide and Peptide bond



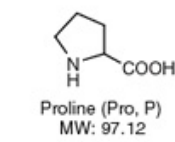
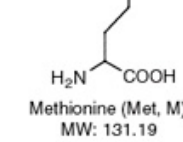
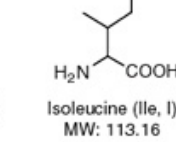
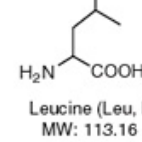
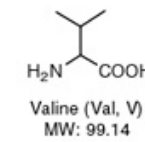
Small



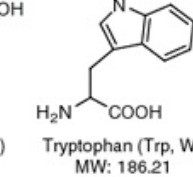
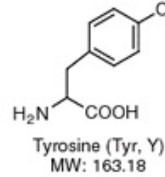
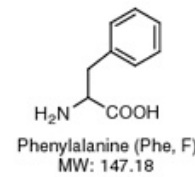
Nucleophilic



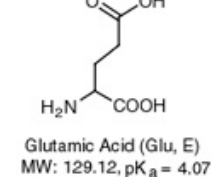
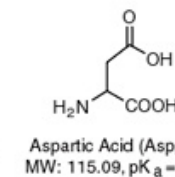
Hydrophobic



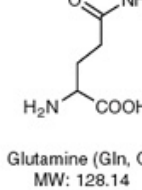
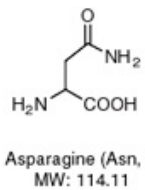
Aromatic



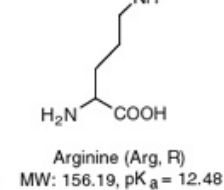
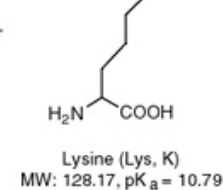
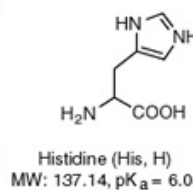
Acidic



Amide



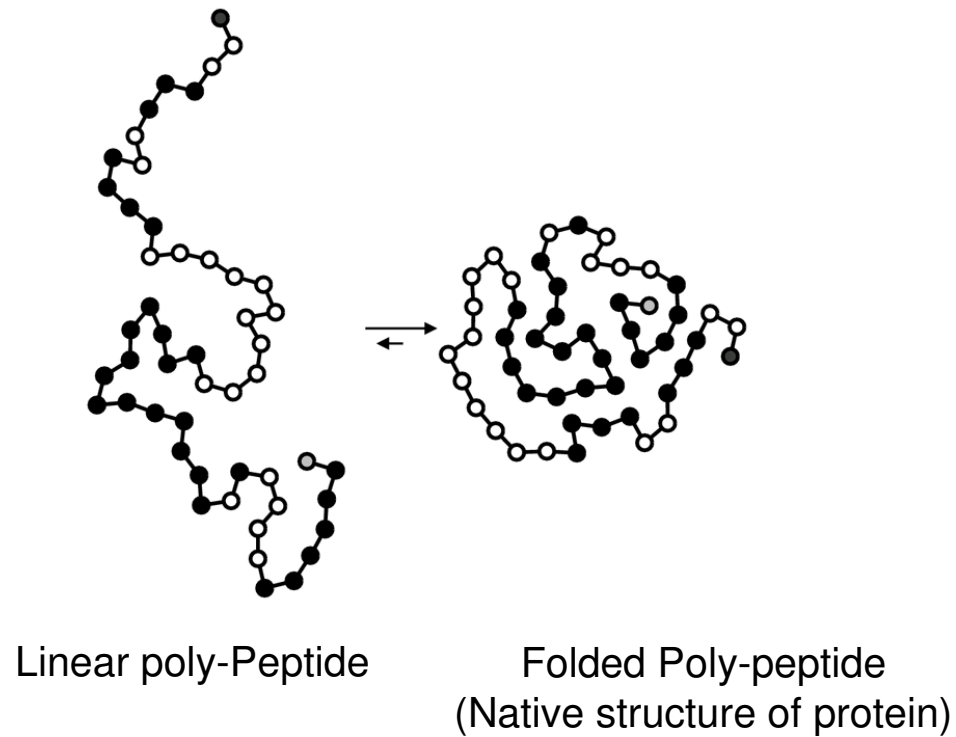
Basic



<https://www.neb.com/tools-and-resources/usage-guidelines/amino-acid-structures>

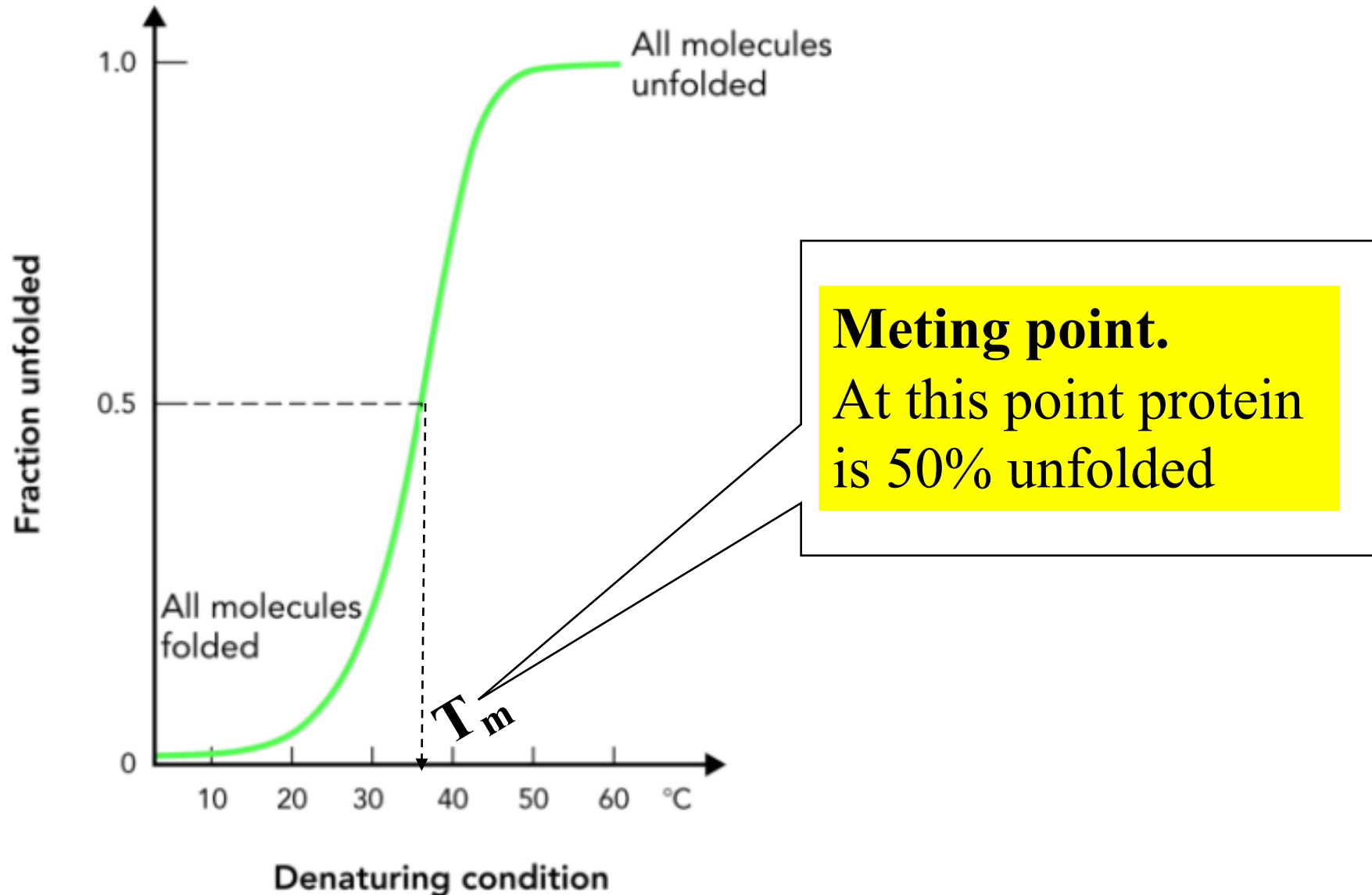
Protein = Chain of amino-acids

Polypeptide (or Protein) Folds in 3D



In water, hydrophobic residues (shown as black circles) tend to hide from water, there by spontaneously folding protein to its native structure.

A protein is functional at its native structure.



How much time is required for folding ?

Lets take a small protein = say 101 aa

Number of peptide bonds = 100

Assumption : **say 3 conformations** are possible around each peptide bond

Total number of minima/structure = $3^{100} = 5 \times 10^{47}$

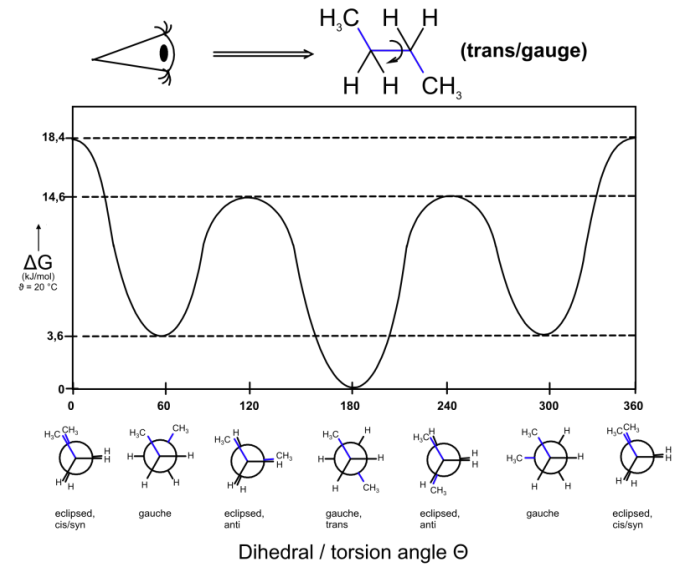
Typical rotations take $\sim 10^{-13}$ Sec

So a search of all conformation will take
 $= (5 \times 10^{47}) \times 10^{-13} = 5 \times 10^{34}$ sec = $(5 \times 10^{34}) / (365 \times 24 \times 3600) = 1.6 \times 10^{27}$ Years

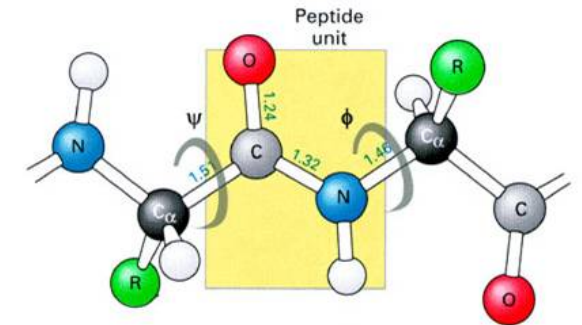
Age of the universe from big bang $\sim 14 \times 10^9$ years

small proteins (300 amino-acids) fold spontaneously on a millisecond or even microsecond time scale.

Levinthal's paradox

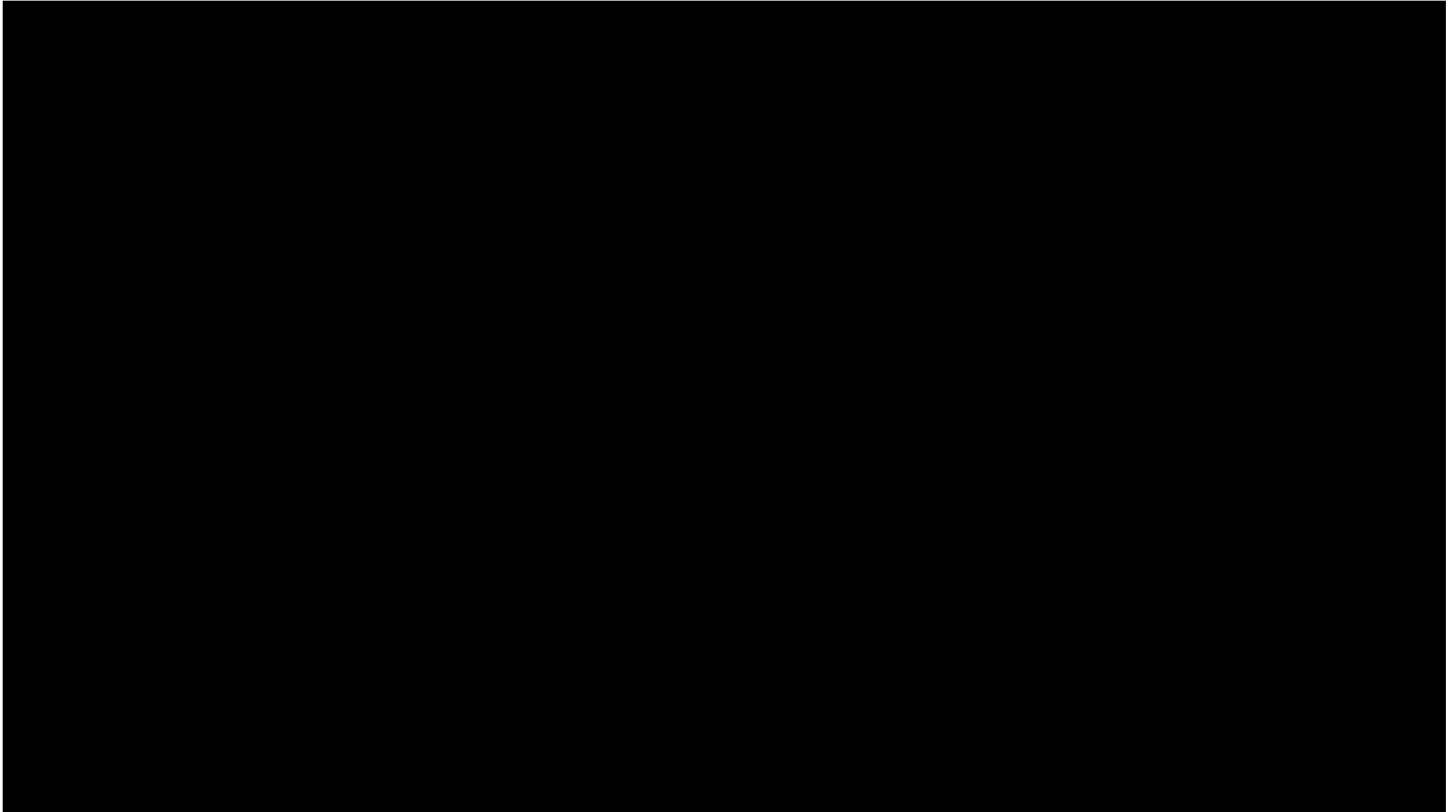


https://en.wikipedia.org/wiki/Dihedral_angle



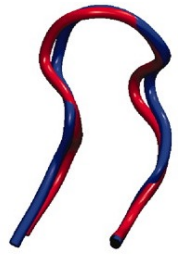
https://www3.cmbi.umcn.nl/wiki/index.php/Phi-psi_angle

Protein-Folding: Is it a Paradox ?

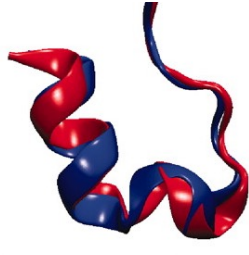


How proteins fold ? <https://www.youtube.com/watch?v=Pjt1Q2ZZVjA>

Good News : We can run simulations with reasonable accuracy and time scale..



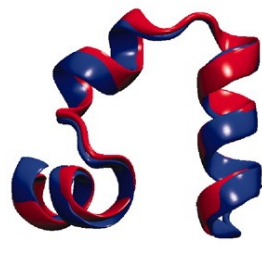
Chignolin 106 μ s
cln025 1.0 Å 0.6 μ s



Trp-cage 208 μ s
2JOF 1.4 Å 14 μ s



BBA 325 μ s
1FME 1.6 Å 18 μ s



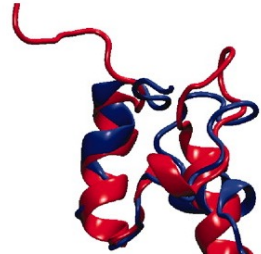
Villin 125 μ s
2F4K 1.3 Å 2.8 μ s



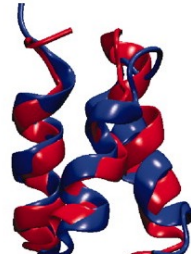
WW domain 1137 μ s
2F21 1.2 Å 21 μ s



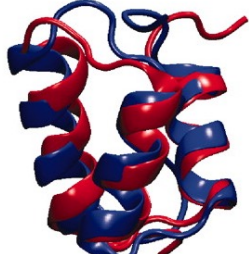
NTL9 2936 μ s
2HBA 0.5 Å 29 μ s



BBL 429 μ s
2WXC 4.8 Å 29 μ s



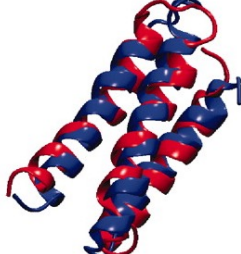
Protein B 104 μ s
1PRB 3.3 Å 3.9 μ s



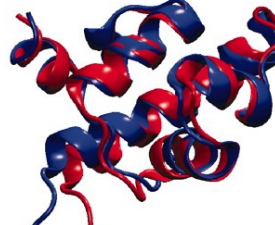
Homeodomain 327 μ s
2P6J 3.6 Å 3.1 μ s



Protein G 1154 μ s
1MIO 1.2 Å 65 μ s



α3D 707 μ s
2A3D 3.1 Å 27 μ s



λ-repressor 643 μ s
1LMB 1.8 Å 49 μ s

Protein-Folding: Is it a Paradox ?

Answer:

Not really.

It is not random.

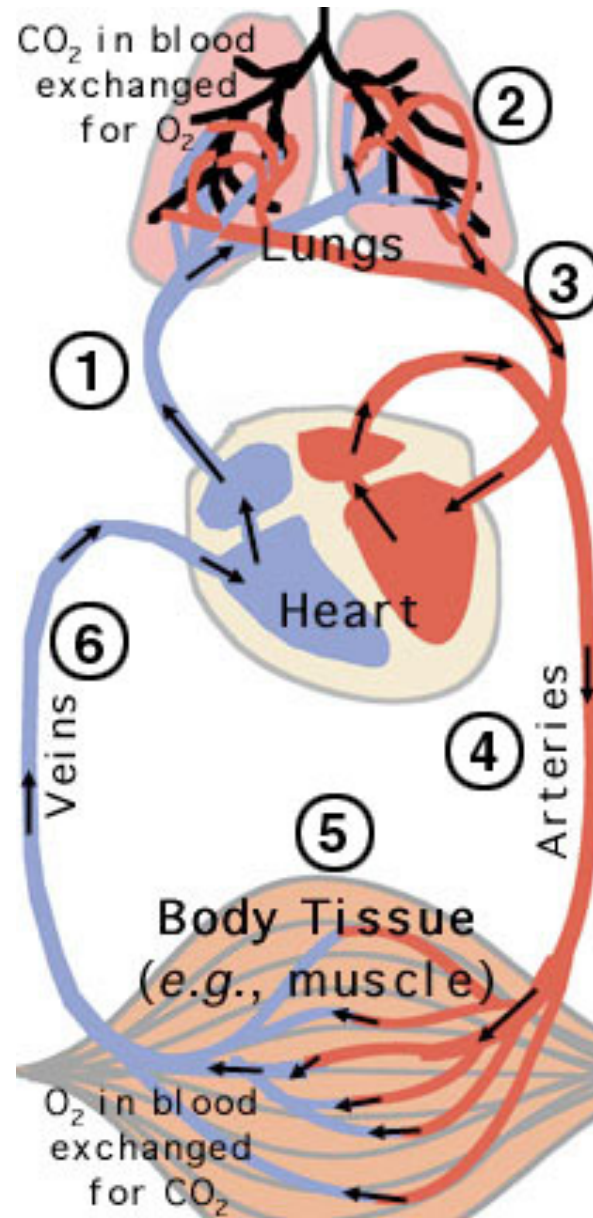
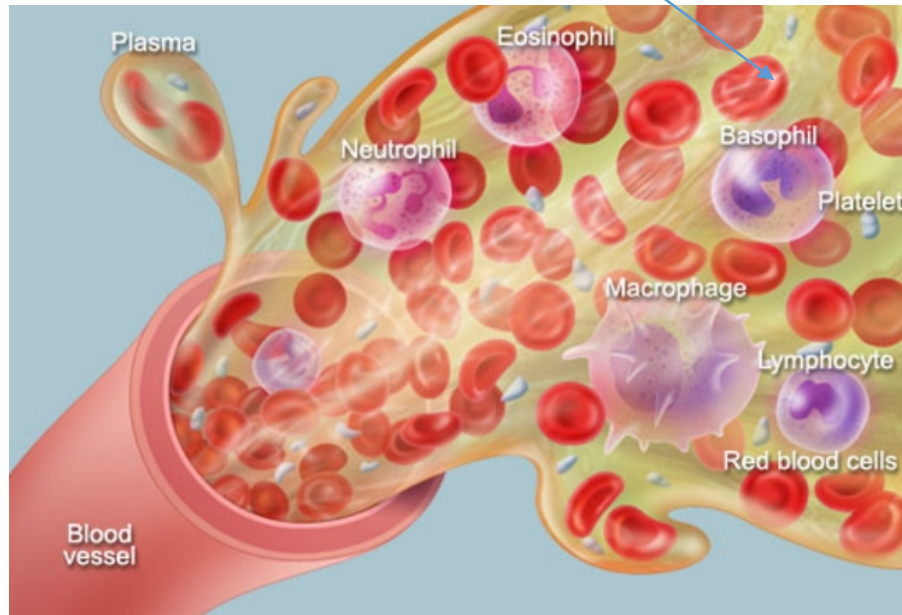
Directed by local and non-local interactions

simulation (blue)

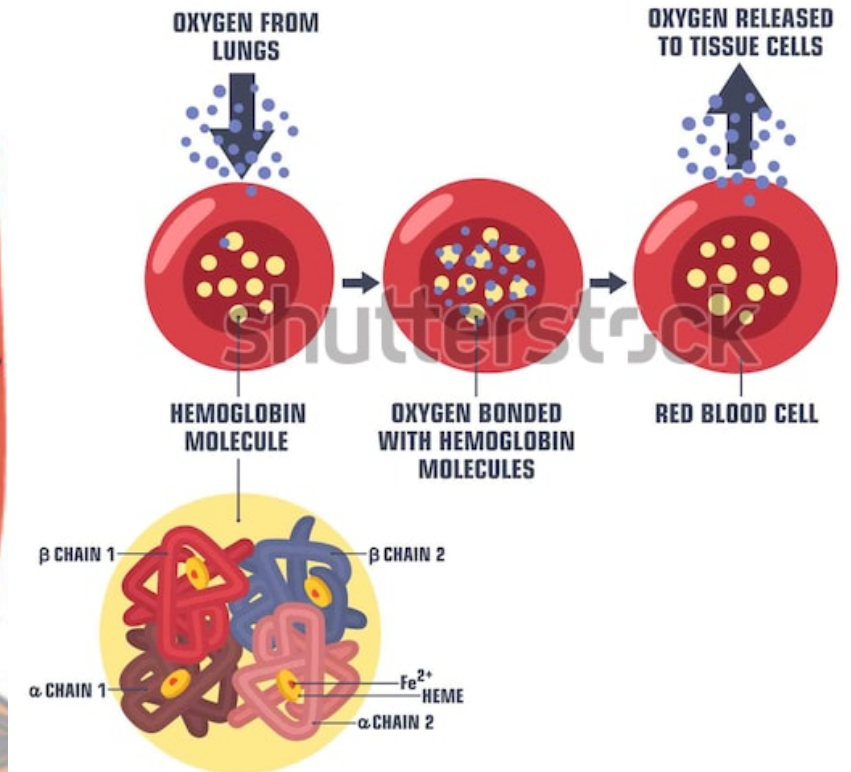
X-ray Structure (red)

Lindorff-Larsen et al.,
Science (2011) 334: 517-520

Proteins present in our blood: Hemoglobin



TRANSPORT OF OXYGEN IN THE BLOOD CELLS



www.shutterstock.com · 1472480087

25 trillion red blood cells (RBCs) circulate in the bloodstream of an adult individual, each one packed with ~260 million haemoglobin molecules.

Hemoglobin: Sickle-cell Disease

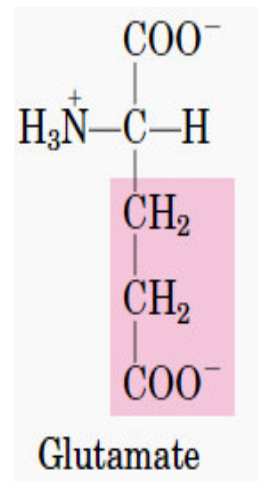
A single Base Mutation:

Normal beta chain

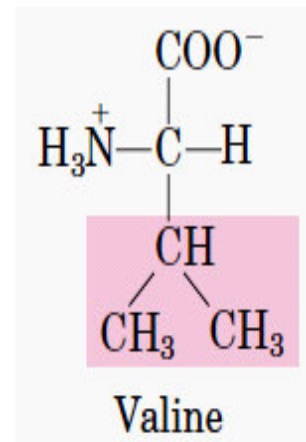
ATG GTG CAC CTG ACT CCT GAG GAG AAG TCT GCC GTT ACT
Val His Leu Thr Pro Gln Gln Lys Ser Ala Val Thr

HbS Sickle cell (missense)

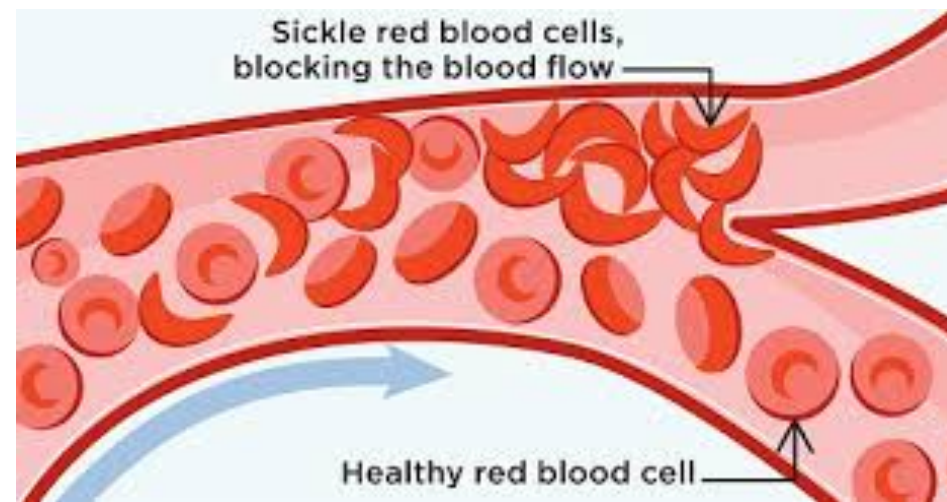
ATG GTG CAC CTG ACT CCT GTG GAG AAG TCT GCC GTT ACT
Val His Leu Thr Pro **Val** Gln Lys Ser Ala Val Thr



Hydrophilic



Hydrophobic

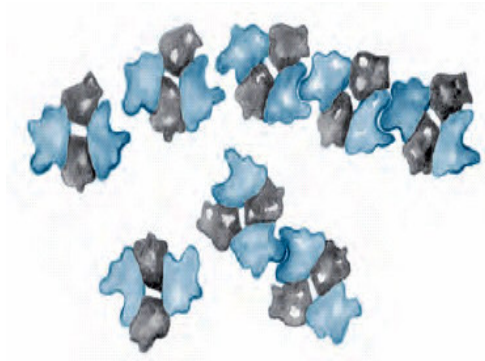


Sickle cell

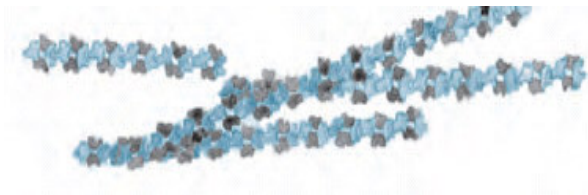
Normal

- RBC changes shape
- Oxygen carrying capacity decrease
- Sickle shaped RBC blocks blood vessels
- Increased lysis of RBC

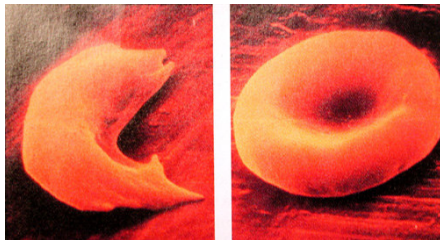
Sickle-cell Disease



- Glu → Val mutation generates a hydrophobic patch in each β subunit.
- In deoxygenated state such hydrophobic patches get exposed.
- Two hemoglobin interacts through those hydrophobic patch and stick.



- Forms elongated strands of sticking hemoglobins
- Crystallize and precipitates

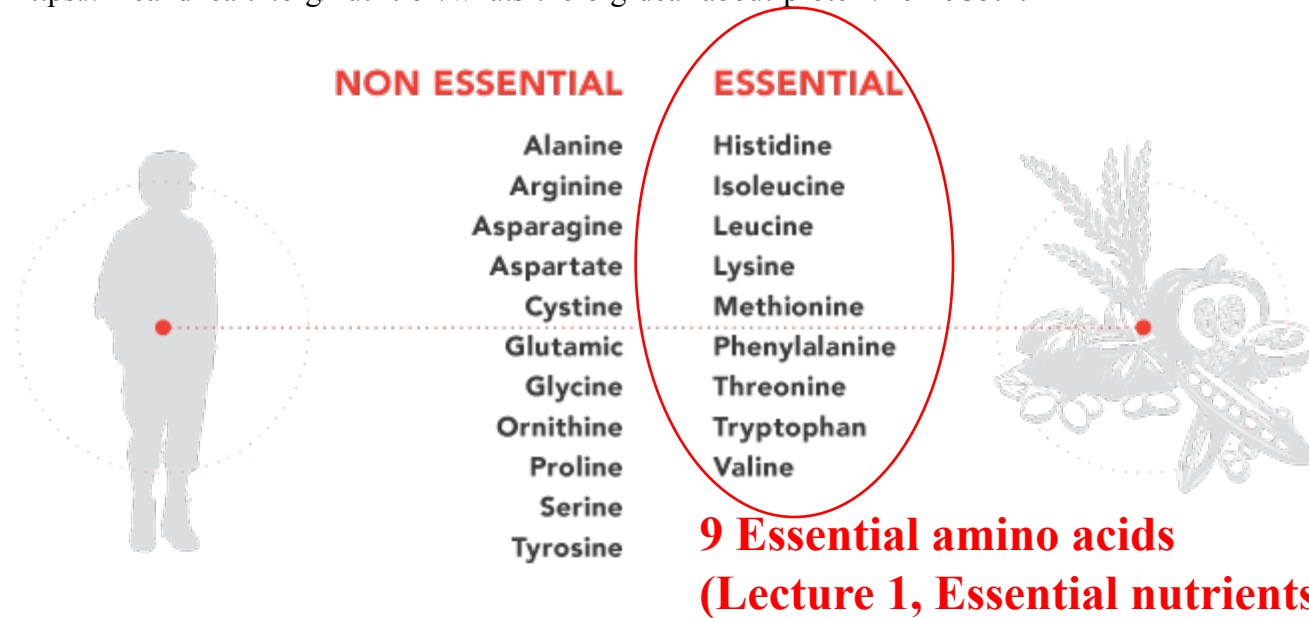


Sickle cell

Normal

- RBC changes shape
- Oxygen carrying capacity decrease
- Sickle shaped RBC blocks blood vessels
- Increased lysis of RBC

Protein Quality



• Protein Quality

- **Complete proteins:** supply all essential amino acids
 - “high-quality proteins”
 - Examples: animal proteins, soy proteins
- **Incomplete proteins:** low in one or more essential amino acids
 - “low quality proteins”
 - Examples: most plant proteins

Amino Acid	Main Food Sources
Histidine	soy protein, eggs, parmesan, sesame, peanuts
Isoleucine	eggs, soy protein & tofu, whitefish, pork, parmesan
Leucine	eggs, soy protein, whitefish, parmesan, sesame
Lysine	eggs, soy protein, whitefish, parmesan, smelts
Methionine	eggs, whitefish, sesame, smelts, soy protein
Cysteine	eggs, soy protein, sesame, mustard seeds, peanuts
Phenylalanine	eggs, soy protein, peanuts, sesame, whitefish
Tyrosine	soy protein, eggs, parmesan, peanuts, sesame
Threonine	eggs, soy protein, whitefish, smelts, sesame
Tryptophan	soy protein, sesame, eggs, winged beans, chia seeds
Valine	eggs, soy protein, parmesan, sesame, beef

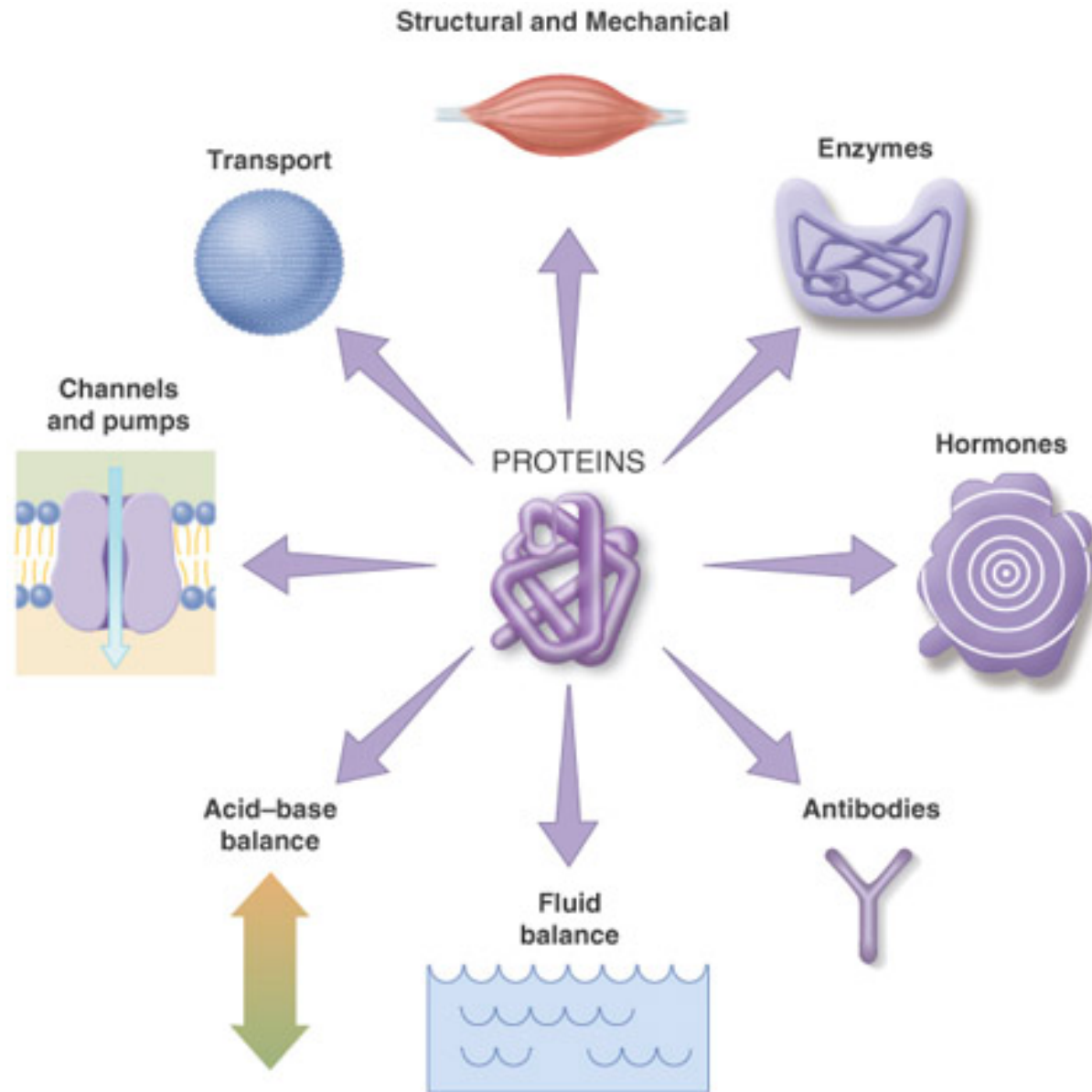
<https://www.healthkart.com/connect/the-all-essential-amino-acids-foods-list-you-must-know-about/bid-6106>

Total daily requirement for protein

Males ~ 56g/day,

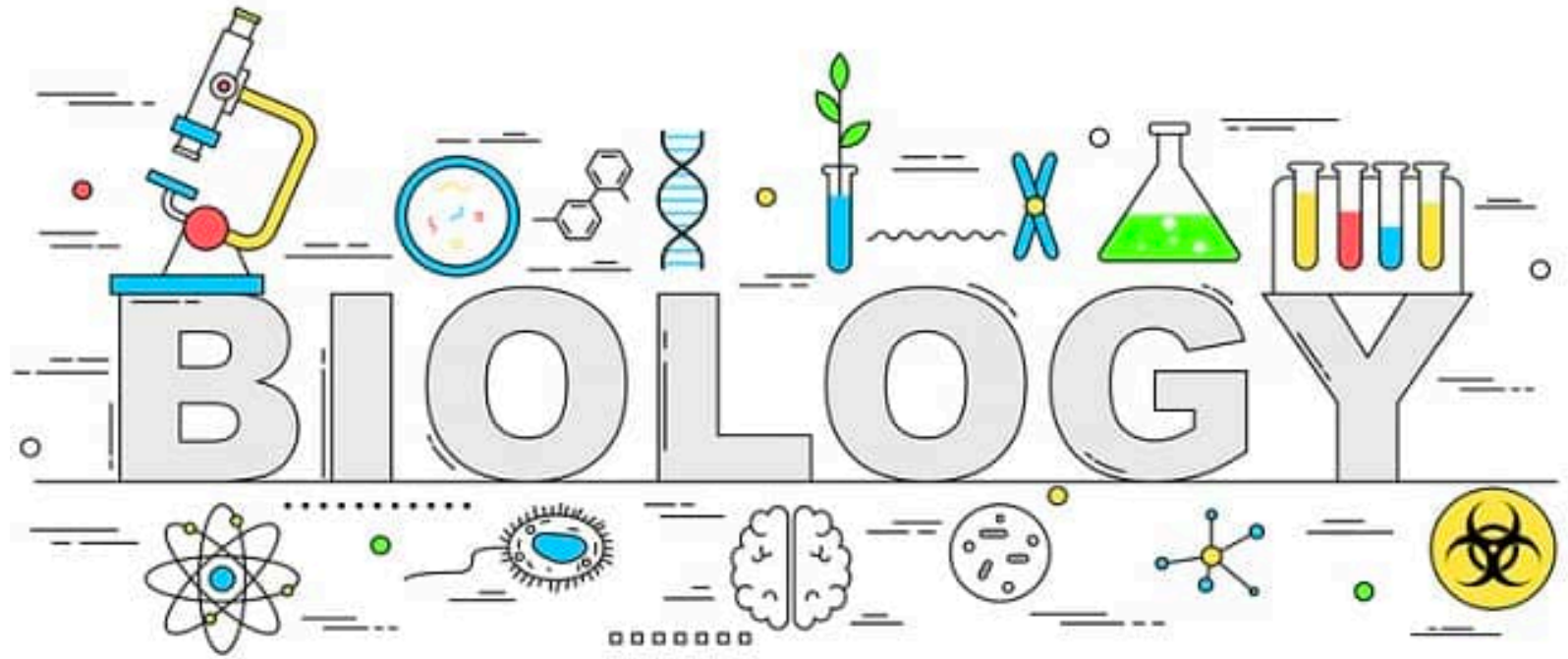
Females ~ 46 g/day

Function of proteins



Take home

- **Proteins** are one or multiple chains of amino acid residues (peptide bonds).
- Protein Folding results in functional protein
- Melting point (T_m) stability of proteins
- Levinthal's paradox (NOT A PARADOX)
- Hemoglobin.
- Mutation causes disease (Sickle cell anemia)
- 9 essential amino-acids



Next: Lipids